## A · Clinical need versus image physics

**1 Regional function is a distinct measurement problem**
Ejection fraction compresses 3-D deformation into a chamber ratio; compensation, cancellation and timing differences are hidden.

§1

**2 Cine carries no material-point identity**
Tagging imprints a pattern and DENSE encodes displacement in phase; bSSFP myocardium is homogeneous, so identity is not in the signal.

§2 · Fig. 2

↓ *Leap 1: physics → estimation*

## B · From physical gap to dependent number

**3 Correspondence is inferred, and the prior selects**
Many deformation fields fit the same images; smoothness, incompressibility, biomechanics or a training distribution picks one.

§2–3 · Fig. 1a · Table 1

**4 A reported value depends on three choices**
Which estimator inferred it, how strain was defined, what it was validated against: ‘is cine FT regional strain accurate’ is incomplete.

§2

## C · Evidence audit: what each endpoint supports

**5 Boundary endpoints cannot confirm dense correspondence**
Dice, surface distance, field regularity and sparse landmarks measure boundary geometry and sampled points; identical borders admit different interior motion.

§4 · Table 2

↓ *Leap 2: reported endpoint → supported claim*

**6 Each validation target supports one claim and leaves one gap**
Known motion, paired DENSE or tagging, scan–rescan, LGE, outcome; software, sampling, strain convention, statistical unit and reference uncertainty are stated before attribution.

§4–5 · Fig. 1b · Fig. 3

**7 Paired studies support a threefold qualification**
Circumferential strain, global or whole-slice mid-ventricular scale, matched definitions; agreement varies with component, scale and software, radial least.

§4 · Tables 2–3

**8 Focal attributes are rarely measured directly**
Ranked by what already measures them: location and extent compete with LGE, magnitude has no direct competitor, timing has none and the least evidence.

Fig. 4 · Table 3 · Fig. 5c

## D · Engineering the claim

**9 A measurement contract makes the claim falsifiable**
Acquisition, estimator, population, spatial unit, strain definition and reference state, time point, comparator, prespecified tolerance.

§6

**10 No single agreement index validates all of it**
Known motion: capacity and technical error. Paired material reference: material-sensitive agreement. Scan–rescan: precision. Tissue and decisions: relevance and impact.

§6 · Fig. 5 · Table 4